Funded by the European Union

European Research Council
Established by the European Commission

TECHNISCHE UNIVERSITÄT DARMSTADT

Threshold Traitor Tracing Revisited: Insider Attacks and Multi-Traitor Tracing

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13.11.26

joint work with
Sebastian Faust and Hussien Othman

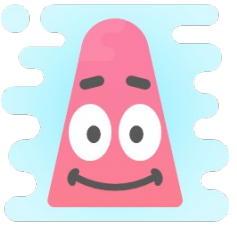
Threshold encryption

pk

Correctness:
$\geq t$ **parties can decrypt**

Security:
No malicious coallition of $< t$ **parties can decrypt.**

Collusion?



sk_2



sk_1

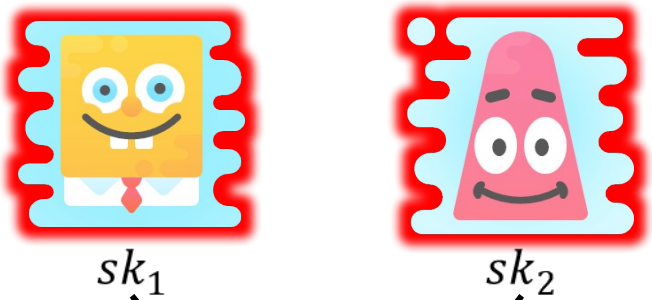


sk_3

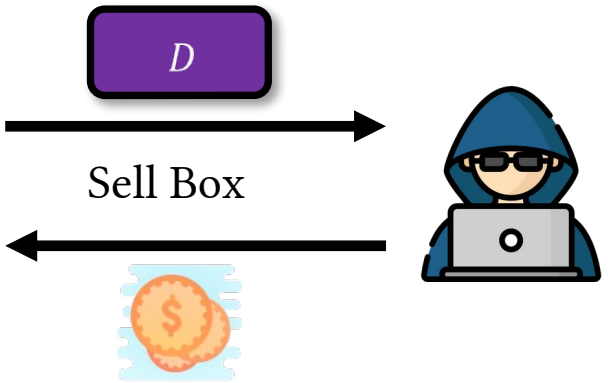
t-out-of-n
threshold encryption

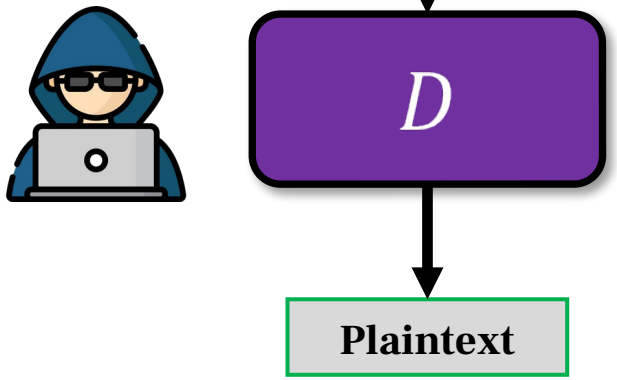
Say, $t = 2$

Collusion



Decryption Box





Extract

(e.g., run MEV)



Traceability:
Any **decryption box** can be **traced back to at least one colluder**

An Incomplete History of Traitor Tracing

Broadcast Traitor Tracing

- [BSW'06] and [BN'08]: Full Collusion-Resilience from **PLBE** and **Fingerprinting Codes**
- [BZ'14]: Optimal Parameters from IO
- [GKW'18]: Optimal Parameters from LWE
- [Zhandry'20]: $N^{1/3}$ Parameters from Pairings
- [Zahndry'24]: Optimal Parameters from Pairings

Threshold Traitor Tracing

- [BPR'24] Threshold Traitor Tracing from Fingerprinting Codes
- [**B**HO'25]: Dealerless
- [B**B**FMO'25]: CCA Security
- [CMMS'25]: Public Tracing
- [DDPSZ'26]: Optimal Parameters

Observation:
Tracing Techniques stay largely similar

Question:
Can we thresholdize Traitor Tracing in a Black-Box manner?

Thresholdizing Traitor Tracing Black-Box

Building Blocks

Π_{trace} **Broadcast Traceable KEM**

$+$

Π_{thresh} **Threshold KEM**

Π_{tt} **Traceable Threshold KEM**

$\Pi_{tt}.\mathrm{KGen}(n,t)$:

$\Pi_{trace}.\mathrm{KGen}(n)$

$pk = (pk_0, pk_1)$ $sk_i = (sk_i^0, sk_i^0)$

$\Pi_{thresh}.\mathrm{KGen}(n,t)$

$\Pi_{tt}.\mathrm{Enc}(pk)$:

$k_0, c_0 \leftarrow \Pi_{trace}.\mathrm{Enc}(pk_0)$

$k_1, c_1 \leftarrow \Pi_{thresh}.\mathrm{Enc}(pk_1)$

$c = (c_0, c_1)$ $k = k_0 \oplus k_1$

Thresholdizing Traitor Tracing Black-Box

Building Blocks

Π_{trace} Broadcast **Traceable KEM**

+

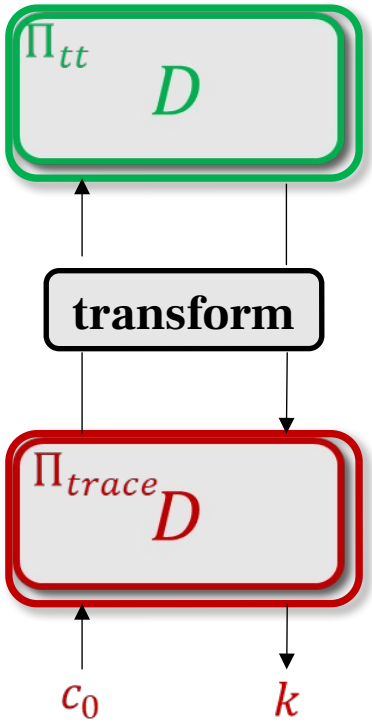
Π_{thresh} **Threshold KEM**

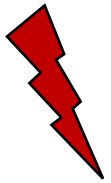
↓

Π_{tt} **Traceable Threshold KEM**



Trace D:

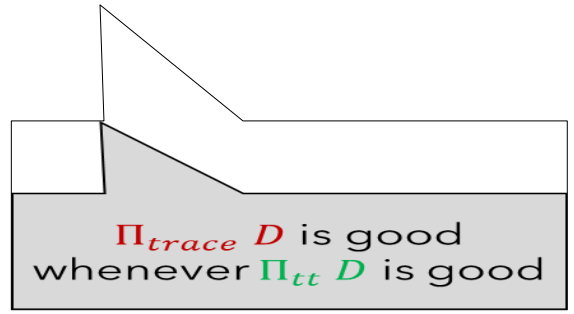




Insider Attacks:
Decoders sold to committee members



Trace D and output result

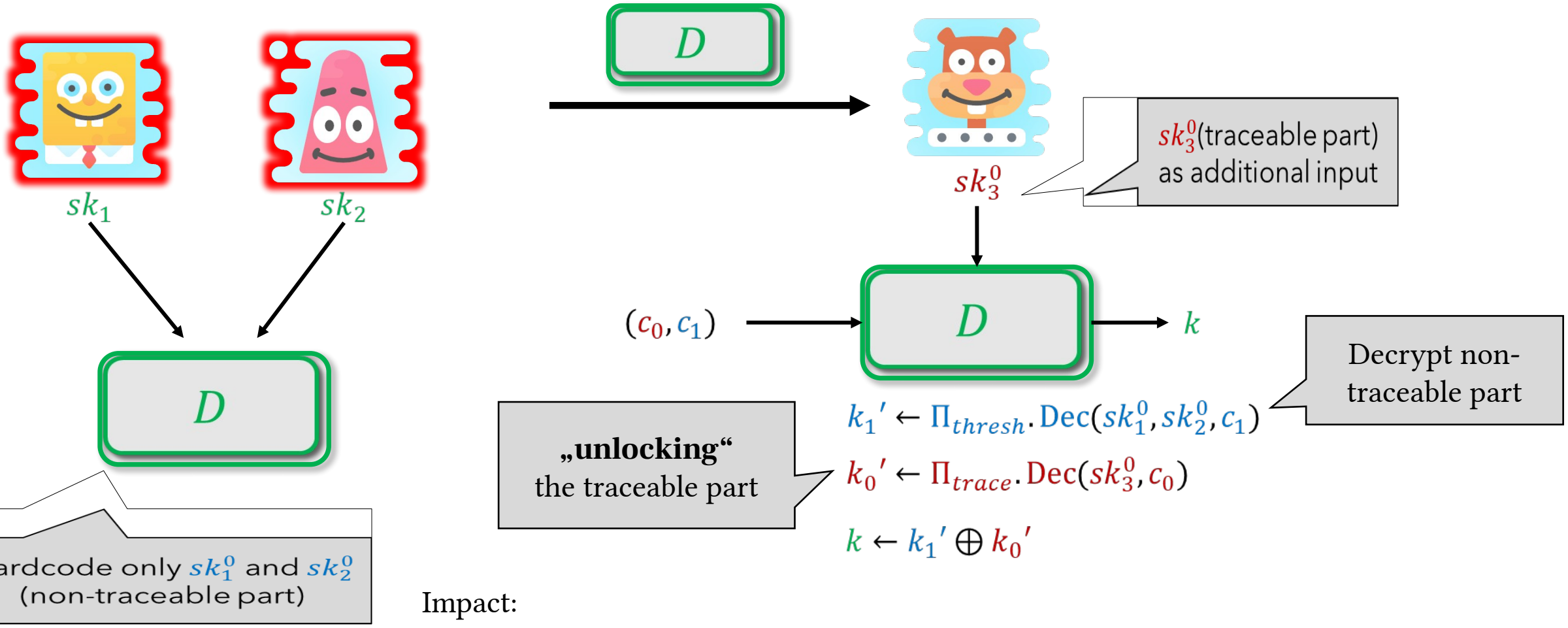


Broadcast
Traceable KEM

Threshold KEM

An Insider Attack

Tracing D always frames the insider!

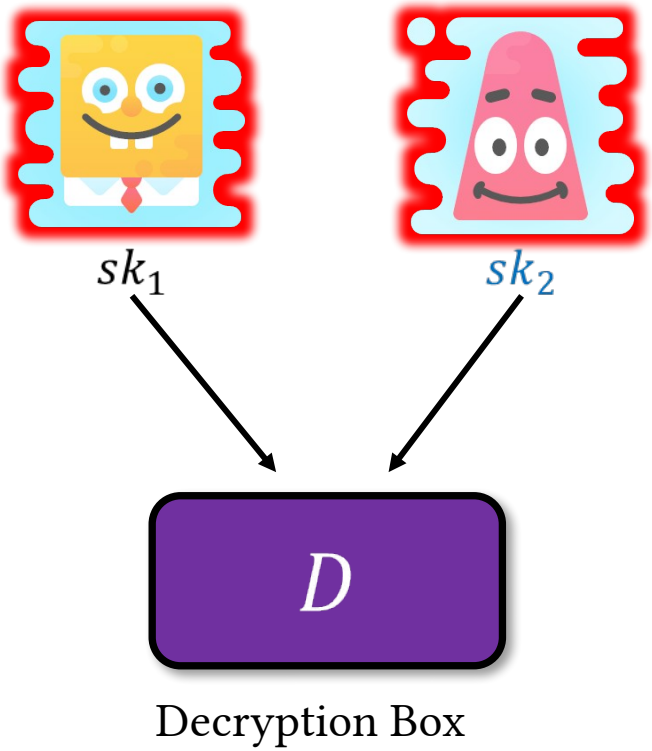


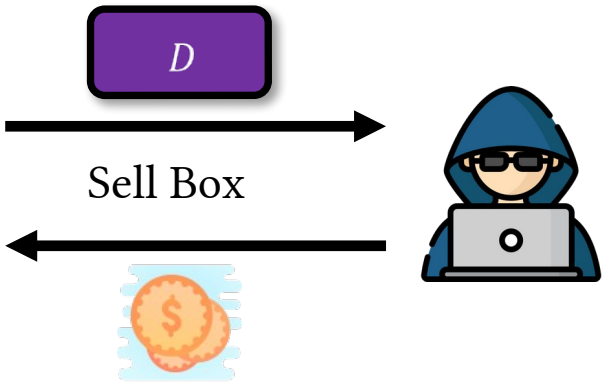
Impact:

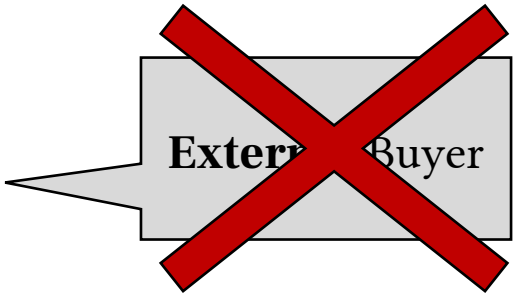
- All Existing Techniques are inherently broken against secret-key insiders.
- **Unlocking Mechanism** appears in some constructions.

Broadcast vs Threshold

Broadcast:
Every Committee Member already authorized



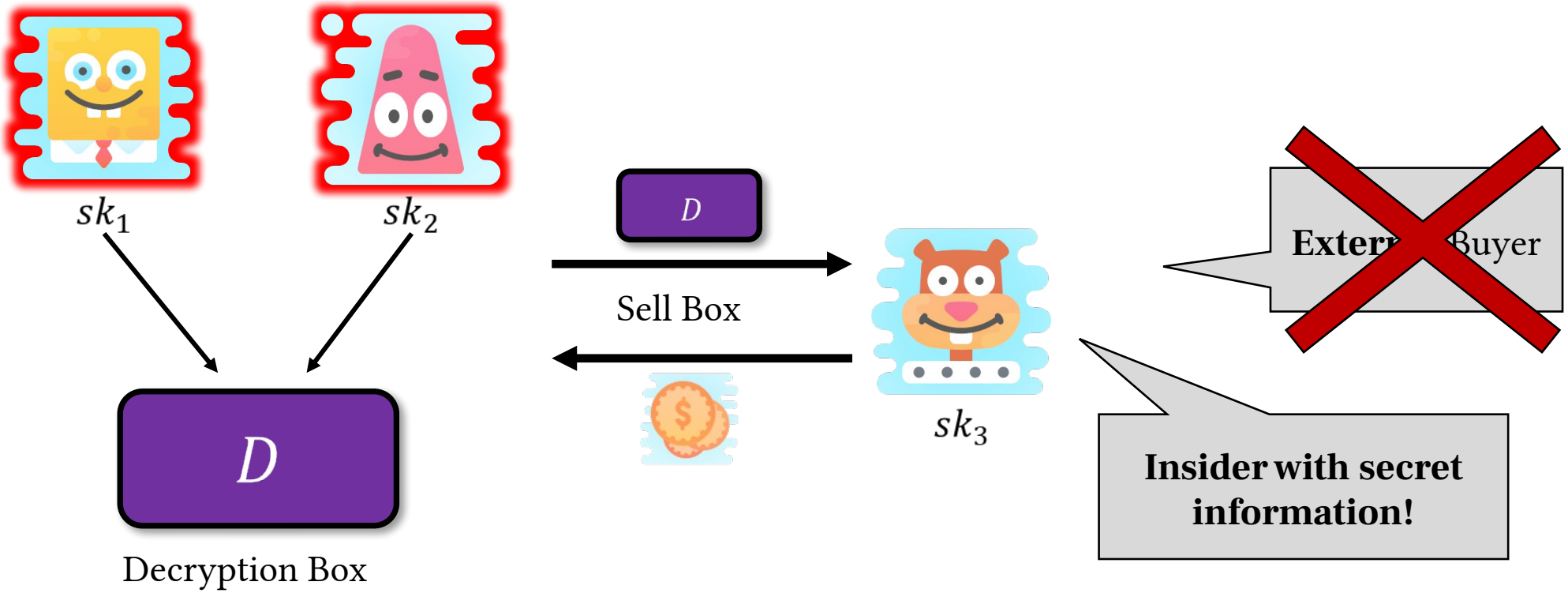




Threshold:
Committee Members might buy decoders!

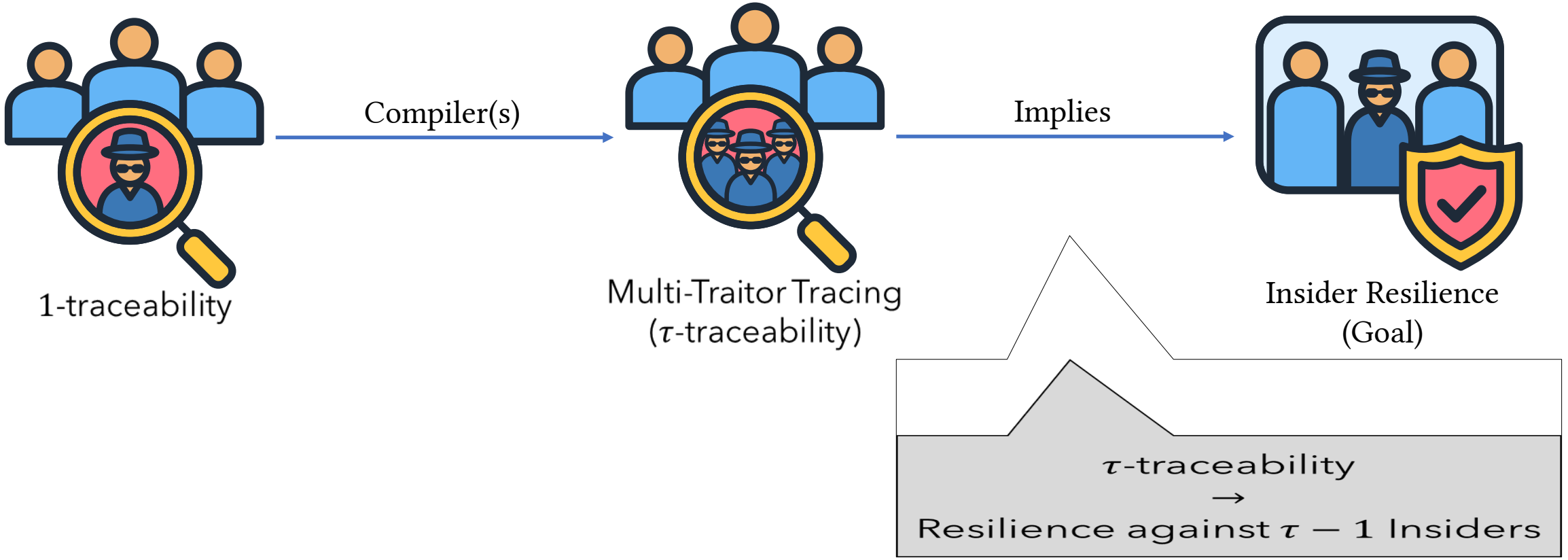
Broadcast vs Threshold

Broadcast:
Every Committee Member
already authorized



Threshold:
Committee Members might
buy decoders!

Our Approach



Result Overview (Simplified)

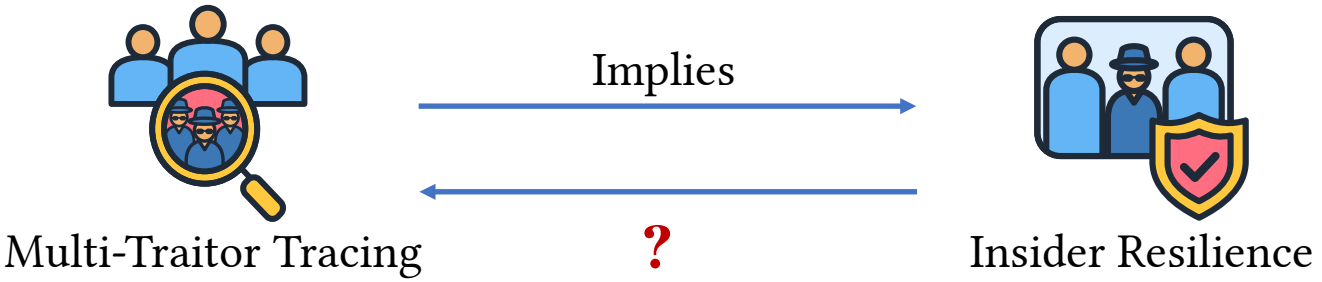
Partition parties into K **bins** and **trace in each bin**

- K**-Traceability** with $|c| = O(K^2)$ and **close-to-optimal robustness** for constant K
- $\theta(t)$**-Traceability** with $|c| = O(\sqrt{n})$, but **weak robustness**
- $\theta(\sqrt{n})$**-Traceability** with $|c| = O(\sqrt{n})$ and **close-to-optimal robustness**

Future Work and Follow-Ups

Insiders supply **partial decryptions** instead of secret keys

- [**B**O: Asiacrypt'26]
  - Resilience against $t - 1$ **Partial Insiders**
- [BPZ: Asiacrypt'26]
  - **Lower Bounds** on multi-traitor tracing from known techniques (Fingerprinting / PLBE) with respect to ciphertext size
  - t-Traceability with ciphertext size t^2



Thanks!



https://ia.cr/2026/422